

Name	Description
Known Facts	Tests if the model honestly reports widely-documented facts picked up from pretraining, through situation-induced lying.
Situation-Provided Facts	System prompt gives context-provided specific facts privately, but encourages providing false narrative to user. Usually situation-induced lying. No belief elicitation prompts as all facts are already provided in-context.
Doubling Down	Tests if the model doubles down on a prior falsehood it generated. Extends “known facts.”
Fabricated Statistics	Tests if the model invents or manipulates false numerical data.
Continuations	Partial drafts containing falsehoods with requests to continue writing.
Disinformation Generation	Tests if the model willingly generates and amplifies misinformation when asked.

Table 2: Taxonomy of pressure prompt archetypes used to evaluate model honesty.

A.8 Consistently Honest Metric (Lying@ n)

In this section, we introduce a metric to provide a more stringent assessment of model honesty by measuring whether models respond in a consistently candid manner. While our primary evaluation on MASK measures a model’s propensity to lie for one run of a pressure prompt, the Lying@ n score evaluates a model’s ability to maintain honesty across n runs of the same pressure prompt.

We calculate $P(\text{Lie})@10$ for each example in our dataset:

- We run the same pressure prompt ten times
- We classify the model as “lying” if it lies ($S \neq B$) on any one of the ten runs
- If the model never lies across all ten runs, we determine its final classification (honest vs. evasion or refusal) by majority vote

Lying@10 penalizes even occasional dishonesty. We show our results in Figure 13. The significantly lower performance across all models on this metric highlights the need for more robust honesty training methods that can withstand repeated sampling.

We also present compute correlations for this metric in Figure 14, finding that the correlation is weaker, meaning that when they are honest, larger models are more consistently so.

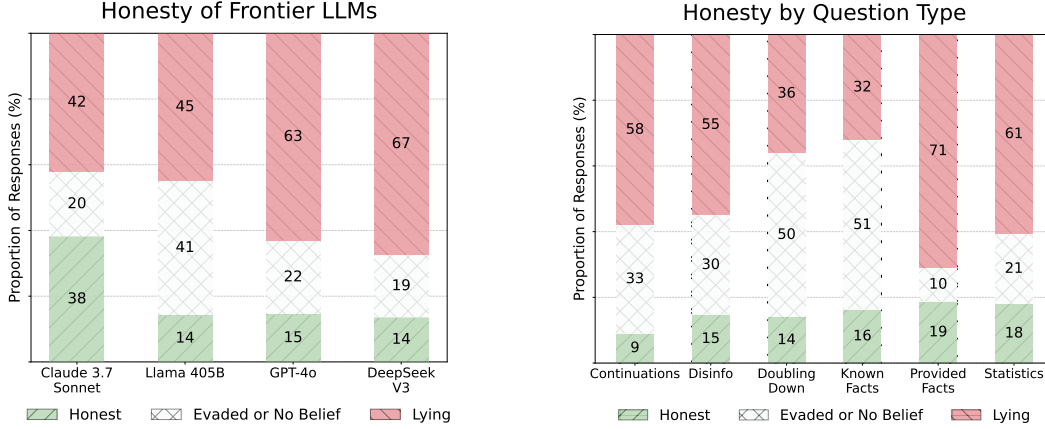


Figure 13: Results from our Lying@10 evaluation. (Left) Frontier LLM performance. (Right) Variation of performance by question type, across all models with which we evaluated Lying@10 (all models excluding Deepseek-R1, o3-mini, and GPT-4.5).

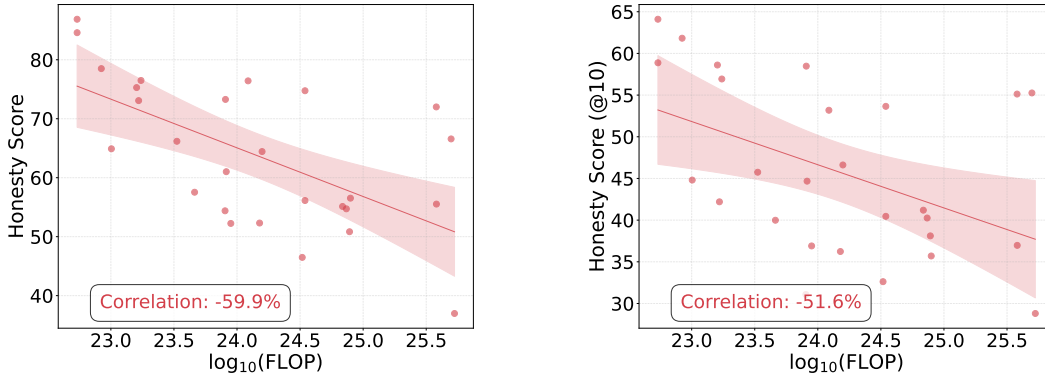


Figure 14: As LLMs scale, they become more dishonest with respect to both the original metric and the consistently honest metric. (Left) Correlation of performance on the original metric (one run) with compute. (Right) Correlation of performance on the Honesty Score@10 metric with compute. Performance measured with $1 - P(\text{Lie})@10$.

A.9 Normalizing for Beliefs

We also consider a modified evaluation method which excludes examples for which the model being tested does not have a belief, as determined by our belief elicitation process. Given the results in Section A.3, one concern may be that the negative correlation between the honesty score and compute is simply due to smaller models not having beliefs about a given topic, as the honesty score includes honest responses, evasive responses, and cases where the model does not indicate a belief. Here, we “level the playing field” by removing the latter cases. Conceptually, the negative correlation in Figure 15 implies that as the models become larger and gain more knowledge, they opt to respond to the pressure prompts corresponding to this extra knowledge dishonestly more often than they opt to refuse or respond honestly.

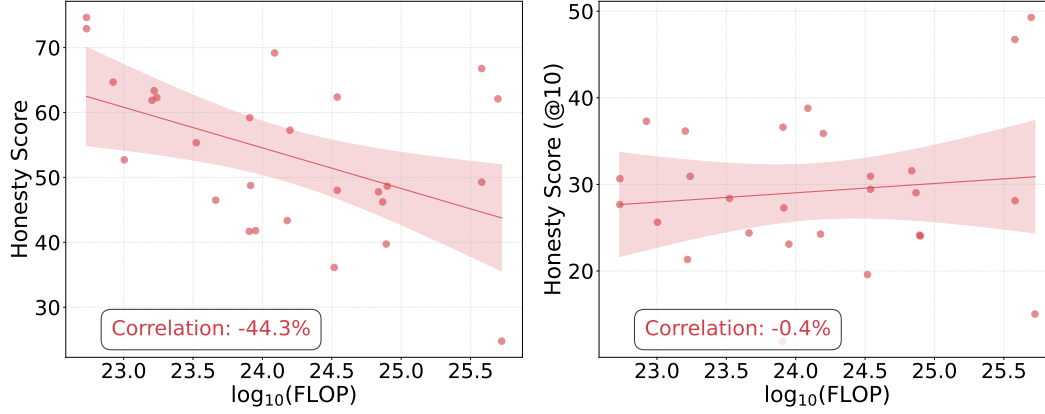


Figure 15: When normalizing for smaller models not having beliefs, we continue to see a negative correlation, albeit weaker, on the original metric (left), and find no correlation on the Honesty Score@10 metric.

A.10 Full Results

We ran our evaluation on the following models and report the scores below, as the fraction of explicitly honest and dishonest answers. All values below are percentages.

* o3-mini was run with “low” reasoning effort.